

A subquadratic upper bound for the mixing time of the Thorp shuffle

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Abstract

Let $n = 2^d$. We prove that the worst-start total-variation mixing time of the full labelled Thorp shuffle satisfies, for every fixed $0 < \varepsilon < 1$,

$$t_{\text{mix}}(\varepsilon) \leq C_* d^{5/3} + O_\varepsilon\left(d^{4/3} \log(d+2)\right), \quad C_* = 5.1536951454\dots$$

Time is measured in ordinary Thorp shuffles. This improves the previously published full-deck upper bound $O(d^3)$ of Morris. The geometric starting point is elementary: label the $n = 2^d$ positions by binary strings. One physical Thorp shuffle is a random swap layer on one coordinate matching of the d -dimensional cube, followed by a deterministic cyclic rotation of the coordinates. In a rotating frame that deterministic rotation disappears, so d successive physical shuffles become one *sweep*: one independent random swap layer in each coordinate direction. The proof then combines five ingredients. First, a collision-graph representation and negative dependence give pointwise smoothing for ordered marked cards. Second, an entropy argument is sharpened by moving an endpoint moment up to the critical exponent $\beta_c = \log_2(3/2)$, yielding an entropy-preparation clock $s^2/(2 \log(3/2)) + O(s \log s)$. Third, exact multiplicities in the ordered h -tuple permutation representation amplify marked-card bounds into Fourier operator and energy estimates on low representation levels. Fourth, entropy truncation produces a positive mixture whose regular component retains the marked-card budget. Finally, only two long blocks need this entropy regularization: a shorter third block is used solely for additional low-level smoothing. Optimizing the three block lengths gives the constant above. We state all conventions and reproduce the shuffle-specific analytic estimates in full; the only black-box inputs are standard strong-Rayleigh negative dependence and a standard high-level dimension estimate for representations of S_n .

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1 Introduction and historical context

The Thorp shuffle is a particularly local model of a riffle shuffle. For an even number n of cards, split the deck into two equal packets. For each pair of current bottom cards, toss a fair coin to decide whether the left card or the right card is dropped first, and then drop the other card. Continue until both packets are empty. Edward Thorp introduced this model in 1973 in his study of nonrandom shuffling and the game of Faro [11]. In contrast with the Gilbert–Shannon–Reeds riffle shuffle, whose total-variation behavior was determined sharply by Bayer and Diaconis [2], the Thorp shuffle resisted analysis for decades.

For $n = 2^d$, the first polylogarithmic full-deck upper bound was obtained by Morris: $O(d^{44})$ [7]. Montenegro and Tetali sharpened this to $O(d^{29})$ using spectral-profile/evolving-set methods [6]. Morris subsequently proved an $O((\log n)^4)$ bound for arbitrary even n [8], and then, for $n = 2^d$, reduced the full-deck bound to $O(d^3)$ [10]. In a different direction, Morris, Rogaway and Stegers obtained strong estimates for projected Thorp shuffles, showing that very large proper subsets of the cards mix much faster than the whole deck [9]. Czumaj and Vöcking later proved an $O((\log n)^2)$

statement for linear-size partial permutations; that theorem does not by itself give an $O((\log n)^2)$ full-labelled mixing bound [4]. Recent work on permutation sampling still cites the cubic Thorp bound as the full-deck benchmark [1]. Thus, to the author’s knowledge, the result below is the first full-deck improvement in the exponent since the cubic bound.

For convenience, the history for powers of two is summarized in Table 1. The entries refer to full-deck mixing unless noted.

Work	setting	upper bound / conclusion
Morris [7]	$n = 2^d$	$O(d^{44})$
Montenegro–Tetali [6]	$n = 2^d$	$O(d^{29})$
Morris [8]	even n	$O((\log n)^4)$
Morris [10]	$n = 2^d$	$O(d^3)$
Morris–Rogaway–Stegers [9]	projections	rapid partial mixing
Czumaj–Vöcking [4]	linear-size partial permutations	$O((\log n)^2)$
This paper	$n = 2^d$	$C_* d^{5/3} + O_\varepsilon(d^{4/3} \log d)$

Table 1: Selected milestones for the Thorp shuffle.

There is a simple information-theoretic lower bound of order d . One ordinary Thorp shuffle uses exactly $n/2$ independent fair bits, so after r shuffles a fixed initial deck can have support of size at most $2^{rn/2}$. If its law is within total variation $\varepsilon < 1$ of uniform on S_n , its support must have size at least $(1 - \varepsilon)n!$. Hence

$$r \geq \frac{2}{n} \log_2((1 - \varepsilon)n!) = 2d - 2 \log_2 e + o(1). \quad (1.1)$$

Thus a substantial gap remains between the linear lower bound and the $d^{5/3}$ upper bound proved here.

1.1 Main theorem

Set

$$L = \log 2, \quad A = \frac{1}{2 \log(3/2)}, \quad z_* = \frac{17 + \sqrt{97}}{8},$$

and

$$C_* = \left(2 + \frac{1}{z_* - 2}\right) A^{1/3} (z_* L)^{2/3} = 5.153695145414931 \dots \quad (1.2)$$

All logarithms are natural unless a base is displayed explicitly.

Theorem 1.1 (Main theorem). *Let $n = 2^d$, and let $t_{\text{mix}}(\varepsilon)$ denote the worst-start ε -mixing time in total variation, measured in ordinary Thorp shuffles. For every fixed $0 < \varepsilon < 1$,*

$$t_{\text{mix}}(\varepsilon) \leq C_* d^{5/3} + O_\varepsilon(d^{4/3} \log(d + 2)). \quad (1.3)$$

The proof is quantitative throughout. The value C_* is not asserted to be optimal for the shuffle. It is the exact minimizer within the asymmetric three-block schedule used in Section 8.

1.2 From the physical shuffle to the hypercube, step by step

This subsection spells out the basic geometric model in detail. Nothing from representation theory or entropy is needed here.

The physical procedure. Assume $n = 2^d$. Split the deck into two packets of $n/2$ cards, keeping the order inside each packet. Pair the cards having the same depth in the two packets. For every pair toss an independent fair coin, and use that coin to decide which member of the pair occupies the earlier of the two corresponding output positions. This is one Thorp shuffle. Thus there are $n/2$ independent coin tosses in one shuffle.

The binary model records exactly the same operation. The binary strings label *positions*, not card identities. Label the n positions by the vertices of the cube

$$Q_d = \mathbb{F}_2^d = \{0, 1\}^d.$$

Write a vertex as (b, x) , where $b \in \mathbb{F}_2$ is the first bit and $x \in \mathbb{F}_2^{d-1}$ contains the remaining $d - 1$ bits. The two packets are $b = 0$ and $b = 1$. Positions $(0, x)$ and $(1, x)$ form one pair. For each x , let $\xi_x \in \{0, 1\}$ be the fair coin assigned to that pair. Before any deterministic relabelling, the random part of the shuffle is

$$L_\xi(b, x) = (b \oplus \xi_x, x), \quad (1.4)$$

where $\oplus$ is addition modulo two. If $\xi_x = 0$ the two positions keep their order; if $\xi_x = 1$ they are exchanged. Hence L_ξ is simply a collection of independent random swaps on all cube edges parallel to the first coordinate.

The physical output order moves the packet bit from the front of the binary address to the back. Define the deterministic cyclic rotation

$$R(b_1, b_2, \dots, b_d) = (b_2, \dots, b_d, b_1). \quad (1.5)$$

One physical Thorp shuffle is therefore the position permutation

$$T_\xi = R \circ L_\xi, \quad (b, x) \mapsto (x, b \oplus \xi_x). \quad (1.6)$$

Equation (1.6) is not a different shuffle; it is just the usual two-packet procedure written in binary coordinates.

A concrete eight-card example. Take $d = 3$. The eight positions are

$$000, 001, 010, 011, 100, 101, 110, 111.$$

The first random layer pairs

$$000 \leftrightarrow 100, \quad 001 \leftrightarrow 101, \quad 010 \leftrightarrow 110, \quad 011 \leftrightarrow 111.$$

Consider the card currently at $101 = (1, 01)$. Its pair coin is ξ_{01} . If $\xi_{01} = 0$, the random layer leaves the card at 101 and the rotation sends it to 011. If $\xi_{01} = 1$, the random layer moves it to 001 and the rotation sends it to 010. Thus the compact formula $(b, x) \mapsto (x, b \oplus \xi_x)$ literally describes the physical operation.

Removing the deterministic rotation. The rotation R does not create or destroy randomness. It only renames the positions, so applying a deterministic power of R to every configuration preserves total-variation distance from the uniform distribution. We may therefore observe the chain in a *rotating frame*.

Let $L^{(i)}$ denote a fresh random swap layer on all edges parallel to coordinate i of the cube; the swaps on different edges are independent. If the first physical shuffle uses L_{ξ_1} , the second L_{ξ_2} , and so on, then after r physical shuffles

$$R^{-r}T_{\xi_r}\cdots T_{\xi_1} = (R^{-(r-1)}L_{\xi_r}R^{r-1})\cdots(R^{-1}L_{\xi_2}R)L_{\xi_1}. \quad (1.7)$$

Conjugating by R merely changes which coordinate is called “first.” Consequently the factors on the right of (1.7) are independent random swap layers whose coordinate directions cycle through $1, 2, \dots, d, 1, 2, \dots$

Definition 1.2 (Coordinate layer and sweep). A *coordinate- i layer* independently swaps, with probability $1/2$, the two endpoints of every cube edge parallel to coordinate i . A *sweep* is one coordinate layer in each of the d directions, in a fixed cyclic order.

Since R^d is the identity, a block of exactly d physical Thorp shuffles is not merely comparable to a sweep: after the rotating-frame cancellation it is exactly one sweep, with independent fresh switch bits in all d coordinate layers. Thus

$$\boxed{\text{one sweep} = d \text{ ordinary Thorp shuffles.}} \quad (1.8)$$

All analytic estimates below are stated in sweeps. The last step of the proof multiplies the sweep count by d to return to the physical clock.

Why this viewpoint is useful. In the original two-packet description it is difficult to see how several labelled cards can interact over many shuffles. In the cube, each layer is a perfect matching and a card can only cross one edge of the currently active coordinate. Two marked cards interact only when they arrive at the two ends of the same active edge. This makes their interactions countable by a *collision graph*. It also makes the one-card behavior transparent: during a sweep each coordinate of a given card is refreshed by a fair choice exactly once. Conditional on the card’s entire path up to that layer, the switch bit on the edge it now occupies is still a fresh independent fair bit. Inducting over the d coordinate layers shows that the final d bits are independent fair bits. Thus one marked card is exactly uniform on Q_d after one sweep. The difficulty of full-deck mixing is therefore not making one card random; it is destroying the residual dependence among many cards.

For the eight-card example, the rotating-frame directions used by the first three physical shuffles are coordinate 1, then coordinate 2, then coordinate 3. Those three layers form one sweep. The fourth physical shuffle starts the next sweep again at coordinate 1.

2 Notation and elementary facts

A deck arrangement is a permutation of the n labels among the n cube vertices. We identify the state space with the symmetric group S_n ; our composition convention is the one encoded by the convolution formula below. The particular convention is immaterial for mixing because inversion and left/right translation preserve the uniform distribution.

Let U denote uniform probability on S_n . If ν is a probability law on S_n , its density with respect to U is the function

$$f(\sigma) = \frac{\nu(\sigma)}{U(\sigma)} = n! \nu(\sigma).$$

Thus uniformity is represented by the constant function $f \equiv 1$, and $\mathbb{E}_U f = 1$. The advantage of this normalization is that distances from uniformity become ordinary norms of $f - 1$.

For densities f, g we use normalized convolution

$$(f * g)(x) = \mathbb{E}_{y \sim U} f(xy^{-1})g(y),$$

so that convolution of densities corresponds to composing independent random permutations. If q_r is the density of the walk after r complete sweeps started from the identity, then $q_r * q_s = q_{r+s}$. Thus an expression such as $q_u * q_t * q_t$ is the law of three genuine consecutive blocks whose total length is $2t + u$ sweeps.

For a density f define

$$\mathcal{E}(f) = \mathbb{E}_U[f \log f], \quad \|f\|_p = (\mathbb{E}_U|f|^p)^{1/p}.$$

For a random variable taking values in a finite set equipped with a specified uniform law, $\mathcal{E}(X)$ means the relative entropy of its distribution with respect to that uniform law. For distributions p, q on the same finite set, $D(p||q) = \sum_x p(x) \log(p(x)/q(x))$ with the usual conventions. We write $I(X; Y | Z)$ for conditional mutual information.

For two probability measures ν, π on a finite set,

$$\|\nu - \pi\|_{\text{TV}} = \frac{1}{2} \sum_x |\nu(x) - \pi(x)|.$$

If f is the density of ν relative to π , then Cauchy–Schwarz gives

$$\|\nu - \pi\|_{\text{TV}} \leq \frac{1}{2} \|f - 1\|_2. \quad (2.1)$$

If X_r^{phys} denotes the deck after r ordinary Thorp shuffles, define

$$t_{\text{mix}}(\varepsilon) = \min \left\{ r : \max_{\sigma \in S_n} \left\| \mathcal{L}_\sigma(X_r^{\text{phys}}) - U \right\|_{\text{TV}} \leq \varepsilon \right\}.$$

Here $\mathcal{L}_\sigma$ means that the chain starts from arrangement σ . Left or right translation on S_n preserves both U and total variation, so it is enough to analyze the walk started from the identity. We also write $\log^+ x = \max\{\log x, 0\}$.

For ordered distinct h -tuples of vertices let

$$\Omega_h = \{(x_1, \dots, x_h) \in (\mathbb{F}_2^d)^h : x_i \neq x_j \ (i \neq j)\}, \quad N_h = (n)_h.$$

When a permutation law has density f , P_h^f denotes its stochastic transition matrix on Ω_h . For one sweep we abbreviate this matrix to P_h . Hilbert–Schmidt norms of finite matrices are the unnormalized sums

$$\|M\|_{\text{HS}}^2 = \sum_{x,y} |M(x,y)|^2.$$

This convention is important in Section 6.

The proof also uses the standard representation theory of S_n . A *partition* $\lambda \vdash n$ is a nonincreasing list $\lambda_1 \geq \lambda_2 \geq \dots$ of nonnegative integers summing to n . Its Young diagram has λ_i boxes in row i ; the conjugate partition λ' is obtained by reflecting that diagram, so λ'_1 is the first column length. Irreducible representations of S_n are indexed by partitions. We denote the representation by V_λ and its dimension by $f^\lambda = \dim V_\lambda$.

For this paper the important statistic is the *level*

$$\ell(\lambda) = \min\{n - \lambda_1, n - \lambda'_1\}. \quad (2.2)$$

A small level means that the Young diagram is close either to one long row or to one long column. Informally, these are the Fourier modes that can already be detected by following only a small number of labelled cards. Level zero consists of the constant mode and the sign mode. This intuition is made exact in Section 6.

2.1 A short dictionary for first-time readers

The proof moves among several languages. The following dictionary may be useful on a first reading.

Object	Meaning in this proof
$\mathcal{E}(f)$	How much information about the starting arrangement remains beyond a uniform deck. It is zero exactly at uniformity.
h marked cards	A test that ignores all but h specified labels. Small- h tests are much easier to analyze geometrically.
P_h	The transition matrix seen by those h ordered labels during one sweep.
Representation level k	A Fourier scale on S_n . Low levels are controlled by ordered tuples; high levels are controlled using dimension growth.
Truncation $F = pf + (1-p)b$	An exact decomposition of a mildly nonuniform block into a large “regular” part f and a rare exceptional part b .
Long/short blocks	Actual consecutive pieces of the shuffle. Two long blocks pay for entropy regularization; the third only supplies extra low-level contraction.

No step treats these as metaphors: each row is converted below into a precise inequality. The table only indicates why that inequality is being proved.

2.2 Two external inputs

Everything specific to the Thorp shuffle is proved below. We use two standard external facts.

- (E1) **Strong Rayleigh negative dependence.** The only consequence ultimately needed is intuitive: after independently randomizing by transpositions, occupation of one vertex cannot make occupation of a disjoint collection of vertices more likely. In particular, if all one-vertex occupation probabilities are at most θ , then negative association gives

$$\mathbb{P}(B \text{ is entirely occupied}) \leq \theta^{|B|}.$$

For completeness, the standard formal language yielding this fact is as follows. For a probability measure μ on subsets of a finite ground set V , its multiaffine generating polynomial is

$$Z_\mu(z) = \sum_{A \subseteq V} \mu(A) \prod_{v \in A} z_v.$$

A real multivariate polynomial is called *stable* if it is nonzero whenever every variable has strictly positive imaginary part; μ is *strongly Rayleigh* when Z_μ is stable. If τ is a transposition of two ground-set elements, the operation $\mu \mapsto (1 - \theta)\mu + \theta \tau\mu$, $0 \leq \theta \leq 1$, is called partial symmetrization. Partial symmetrization preserves the strongly Rayleigh property, and strongly Rayleigh measures are negatively associated: for increasing functions depending on disjoint sets of occupation indicators, their covariance is nonpositive. These are Theorems 4.20 and 4.9 of Borcea–Brändén–Liggett [3].

- (E2) **High representation levels.** This input says that once f has controlled L^2 norm, Fourier modes of very high level are automatically tiny because the corresponding representations have large dimension. Formally, for a probability density f on S_n and $1 \leq K < n/10$, if T_f is convolution by f and $T_f|_{>K}$ denotes its restriction to the sum of isotypic components of level $> K$, then

$$\|T_f\|_{>K} \leq \|f\|_2 \left(\frac{eK}{n} \right)^{K/2}. \quad (2.3)$$

This follows from the dimension estimate used in the proof of Keevash–Lifshitz [5, Theorem 4.4 and Lemma 4.5]; only the dimension part is used here, not their globalness hypothesis.

3 Overview of the proof

The central difficulty

One card becomes perfectly uniform after a single sweep, but the whole deck is far from uniform because the cards share switches and therefore remain highly dependent. The proof has to pass from “a few cards look random” to “the entire permutation looks random.” The five modules below perform that passage.

It helps to separate two tasks. The *geometric task* is to understand a small set of marked cards moving through the cube. The *global task* is to convert those small-set estimates into a statement about all $n!$ deck arrangements. Collision graphs solve the first task. Entropy, truncation, and representation theory solve the second.

Question	Tool	Output used later
How can h specified cards fail to separate?	Collision graph on the cube	Pointwise bound on the h -card transition kernel
How quickly does a whole deck forget information?	One-sweep entropy loss	A time after which the deck density is not too spiky on average
How do h -card estimates control full-deck Fourier modes?	Ordered-tuple representations	Low-level operator and energy bounds
What about rare very large density values and high Fourier levels?	Entropy truncation plus dimension growth	A large regular component with controlled L^2 norm
How should the actual shuffle time be allocated?	Two long blocks plus one shorter block	The optimized $d^{5/3}$ bound

1. Marked-card smoothing from collision graphs. Fix h labelled cards and temporarily ignore all the others. If their initial and final cube vertices are prescribed, then every marked path through a sweep is forced. Usually the h cards ask for hd independent switch bits. Whenever two marked cards occupy the two ends of the same active cube edge, however, they ask for the *same* switch bit. Such an event is a collision and saves one independent condition. This gives the exact routing weight

$$P_h(x, y) = n^{-h} 2^{C(x, y)}$$

for a feasible route, where $C(x, y)$ is its collision count.

For a fixed future sweep, pull every active matching edge back to the input vertices. The resulting collision graph G is a simple d -regular graph, and the number of collisions made by an occupied marked set A is exactly $e_G(A)$. Its geometry satisfies the deterministic isoperimetric bound

$$e_G(B) \leq \frac{|B|}{2} \log_2 |B|.$$

After one independent preliminary sweep, negative dependence gives $\Pr(B \subseteq A) \leq (h/n)^{|B|}$. Expanding over connected edge sets then controls the exponential moment of $e_G(A)$. Interpolating row and column L^p bounds converts that moment estimate into the pointwise smoothing bound

$$N_h P_h^t(x, y) \leq e^{\alpha h^2}.$$

Here $N_h = (n)_h$ is the number of ordered distinct h -tuples. Since uniform motion on ordered tuples would have kernel $1/N_h$, the normalized quantity $N_h P_h^t(x, y)$ is exactly the pointwise likelihood ratio relative to uniform. Thus $e^{\alpha h^2}$ close to one means that every prescribed h -card endpoint is nearly as likely as under uniformity.

2. A sharper entropy clock. Marked-card smoothing is useful only after the full-deck law is prevented from putting too much mass on a tiny exceptional set of permutations. Relative entropy measures this global spikiness. For an arbitrary input arrangement μ , expose one fresh sweep backwards in input order. At each input j the unexposed endpoint is uniform on an explicit

candidate set. A symmetric partner construction produces a random earlier input whose index is uniform on $\{0, \dots, j-1\}$ after averaging over the fresh sweep.

The construction yields two different lower bounds for the entropy loss Δ in one sweep. One detects square-root/Hellinger separation and the other detects relative entropy from an averaged partner law. They are *separate* lower bounds; the proof never adds them and therefore never spends the same entropy loss twice. The deterministic partner weights W_{jk} have the critical moment exponent

$$\beta_c = \log_2(3/2).$$

Working just below this exponent and using an adjustable scalar interpolation inequality gives the preparation time

$$R_*(\eta me^{-s}) = \frac{s^2}{2 \log(3/2)} + O_\eta(s \log(s+2) + \log(d+2))$$

in sweeps.

3. Ordered-tuple multiplicity amplification. This is the bridge from a few marked cards to the entire permutation. Fourier analysis on S_n decomposes an arbitrary error from uniformity into orthogonal representation levels. A low-level representation occurs many times inside the permutation action on ordered h -tuples. For $\lambda = (n-k, \mu)$ its exact multiplicity is

$$m_h(\lambda) = \binom{h}{k} f^\mu,$$

while $f^\lambda \leq \binom{n}{k} f^\mu$. Therefore a Hilbert–Schmidt budget for the marked-card matrix forces small Fourier operators and energies on every low level. A sign-twisted tuple module gives the symmetric statement for partitions close to a long column. The use of *ordered* tuples is essential for this exact multiplicity.

4. Entropy truncation. Small entropy does not imply a small L^2 norm: a density can still have very rare spikes. We therefore cut off the high-density tail. This is done as an exact probability decomposition, not by changing the Markov chain:

$$F = pf + (1-p)b, \quad p \geq 1 - \eta.$$

The regular part f has controlled L^2 norm and obeys the pointwise domination $f \leq F/p$. Hence every marked-card estimate already proved for the actual block F passes to f at the cost of only the factor p^{-1} . The exceptional component has total mass at most η and will simply be paid for in total variation.

5. Two long regular blocks and one short actual block. Two independent copies of the regular component f are enough to suppress high representation levels. A third block is needed only to contract the remaining low levels, so that block need not pay the entropy-preparation cost. Let its actual density be G . The regular part analyzed at the end is

$$g = G * f * f.$$

If the long-block marked parameter is α and the short-block parameter is β_0 , the low-level geometric ratio is essentially $n\alpha\sqrt{\beta_0 K}$. High levels are small because the two f factors have controlled L^2 norm and high representation dimensions are large.

The scale of the final exponent can already be guessed before optimizing constants. To reach entropy roughly ne^{-s} costs order s^2 sweeps, whereas the marked-card smoothing needed at that scale costs order d/s sweeps. Balancing

$$s^2 \asymp d/s$$

gives $s \asymp d^{1/3}$ and a sweep count of order $d^{2/3}$. Since one sweep is d physical Thorp shuffles by (1.8), the physical time is of order $d^{5/3}$. The quantitative optimization in Section 8 and the following section produces $z_* = (17 + \sqrt{97})/8$ and the explicit constant C_* .

The next sections carry this program out quantitatively. The longer shuffle-specific proofs are included after the main argument so that every asserted input can be reconstructed from this manuscript.

4 Quantitative inputs used in the proof

The first input is the precise marked-card statement produced by the collision analysis. Keep the following interpretation in mind. Under the uniform law on ordered h -tuples, each endpoint has probability $1/N_h$. Hence $N_h P_h^t(x, y)$ is the ratio

$$\frac{\Pr\{\text{the } h \text{ marked cards end at } y \mid x\}}{\Pr\{\text{the same endpoint under a uniform permutation}\}}.$$

A bound by $e^{\alpha h^2}$ says that correlations among the h marked cards can inflate any endpoint probability by at most that factor.

For ordered distinct h -tuples, let $N_h = (n)_h$ and P_h be the one-sweep transition matrix. For an integer $m \geq 2$, define

$$\alpha_{d,m} = \frac{8Ld}{n} (2n)^{1/[2(m-1)]}. \quad (4.1)$$

Corollary 4.1. *If $\alpha_{d,m}h \leq m - 1$, then, for every integer $t \geq 2m$,*

$$N_h P_h^t(x, y) \leq \exp(\alpha_{d,m} h^2). \quad (4.2)$$

The same assertion holds for reverse-coordinate sweeps.

Proof. The smoothing theorem in Appendix C has $a = 4d(h/n)(1 - 2^{-1/(m-1)})(2h)^{1/[2(m-1)]}$ and upper exponent $(m-1)ha/(1-a)$. Since $(m-1)(1 - 2^{-1/(m-1)}) \leq L$, one has $a \leq \alpha_{d,m}h/[2(m-1)] \leq 1/2$. The exponent is therefore at most $\alpha_{d,m}h^2$. Additional stochastic transitions preserve the maximum. The $h = 1$ case also follows directly from one-card uniformity. $\square$

The coarse entropy profile proved in Appendix F is

$$\frac{E}{\Delta} \leq 100 + 12 \log^+(n/E), \quad E > 0. \quad (4.3)$$

We use this only to bootstrap a logarithm below, not for the leading coefficient of the new clock.

5 An entropy clock at the critical endpoint-moment threshold

Before entering the calculation, we summarize the objects that appear in it. Let μ be the current random deck arrangement and let one fresh independent sweep act on it. Write

$$E = \mathcal{E}(\mu), \quad \Delta = \mathcal{E}(\mu) - \mathcal{E}(\text{deck after the fresh sweep}).$$

Thus E is the remaining information relative to a uniform deck and Δ is the amount erased by one sweep.

The detailed exposure argument in Appendix D reveals positions from high index to low index. When $j+1$ labels remain unrevealed, p_j is the conditional distribution of the label occupying position j . The partner construction supplies deterministic weights W_{jk} and the comparison distribution

$$q_j = \sum_{k \leq j} W_{jk} p_k.$$

Two aggregate discrepancies are used:

$$S = \sum_j \mathbb{E} \left[1 - \frac{1}{j+1} \left(\sum_a \sqrt{p_j(a)} \right)^2 \right], \quad \mathcal{C} = \sum_j \mathbb{E} D(p_j \| q_j).$$

The first is a square-root/Hellinger-type defect from conditional uniformity; the second measures how distinguishable position j is from its randomized partner. Appendix D proves separately that $\Delta \geq (\log 2)S$ and $\Delta \geq \mathcal{C}$.

In this section the notation $p_j, q_j, W_{jk}, S, \mathcal{C}$ is exactly that of Appendix D. Its information inequalities give

$$E = \sum_j \mathbb{E} D(p_j \| u_{j+1}), \quad \mathcal{C} \leq \Delta, \quad S \leq \Delta/L. \quad (5.1)$$

Put $\beta_c = \log_2(3/2)$.

Lemma 5.1 (General endpoint moment). *For $0 < \beta < \beta_c$, set*

$$R_j^{(\beta)} = (j+1)^\beta \sum_{k \leq j} W_{jk}^{1+\beta}, \quad C_\beta = \frac{1}{3 - 2^{1+\beta}}.$$

On every conditioning atom,

$$(j+1)^\beta \sum_a q_j(a)^{1+\beta} \leq R_j^{(\beta)}, \quad 1 \leq R_j^{(\beta)} \leq (j+1)^\beta, \quad \sum_{j=1}^{n-1} R_j^{(\beta)} \leq C_\beta n.$$

Proof. The matrix $(p_k(a))$ is doubly stochastic. Jensen followed by summation over a bounds $\sum_a (\sum_k W_{jk} p_k(a))^{1+\beta}$ by $\sum_k W_{jk}^{1+\beta}$. The pointwise bounds follow because W_j is a probability vector.

Let $U_j = \sum_{k \leq j} W_{jk}^{1+\beta}$ and $T_d = \sum_{j < 2^d} U_j$. The exact weight recursion in (D.4) gives

$$T_d = (1 + 2^{-1-\beta}) T_{d-1} + 2^{-1-\beta} 2^{(1-\beta)(d-1)}, \quad T_0 = 1.$$

For $Z_d = T_d / 2^{(1-\beta)d}$ this becomes

$$Z_d = (2^{\beta-1} + 1/4) Z_{d-1} + 1/4.$$

Its multiplier is below one precisely when $\beta < \beta_c$; its fixed point is $C_\beta \geq 1$. Therefore $Z_d \leq C_\beta$. Finally $\sum_j R_j^{(\beta)} \leq n^\beta T_d \leq C_\beta n$. $\square$

Lemma 5.2 (Adjustable scalar interpolation). *Let p, q be distributions on ℓ points and suppose $\ell^\beta \sum_a q(a)^{1+\beta} \leq R$ with $R \geq 1$. For $0 < \delta < 1$ and $\beta > 0$ put*

$$c_\beta = 1 + 1/\beta, \quad a_{\beta,\delta} = \frac{1}{\beta(1-\delta)}, \quad b_{\beta,\delta} = \log(8/\delta) + 2 + c_\beta.$$

Writing $V = 1 - \ell^{-1}(\sum_a \sqrt{p(a)})^2$, one has

$$D(p\|u_\ell) \leq c_\beta D(p\|q) + [a_{\beta,\delta} \log(R/V) + b_{\beta,\delta}]V. \quad (5.2)$$

The last term is defined to be zero at $V = 0$.

Proof. If $D(p\|q) = \infty$ there is nothing to prove. Set $f = \ell p$, $A_0 = \mathbb{E}_{u_\ell} \sqrt{f} = \sqrt{1-V}$ and $h(x) = x \log x - x + 1$. As in Appendix E,

$$D(p\|u_\ell) \leq \mathbb{E}_{u_\ell} A_0^2 h(f/A_0^2).$$

Suppose first that $0 < V \leq 1 - \delta$, so $A_0^2 \geq \delta$. Choose $M = 8(R/V)^{1/\beta}$. The elementary scalar truncation Lemma E.1, with $T = M/A_0^2 \geq 4$, gives

$$D(p\|u_\ell) \leq (\log M - \log A_0^2 + 2)V + T_1, \quad T_1 = \sum_{f(a) > M} p(a) \log(f(a)/M).$$

Let $\theta = \beta/(1+\beta)$. The entropy variational inequality gives

$$\theta T_1 \leq D(p\|q) + \log \mathbb{E}_q \exp\{\theta \log^+(f/M)\}.$$

Hölder yields

$$\mathbb{E}_q \exp\{\theta \log^+(f/M)\} \leq 1 + M^{-\theta} R^{1/(1+\beta)} p\{f > M\}^\theta.$$

For $f > M \geq 8$, $f \leq 4(\sqrt{f} - 1)^2$. Hence $p\{f > M\} \leq 8(1 - A_0) \leq 8V$. The added term is at most V by the choice of M . Thus $T_1 \leq c_\beta D(p\|q) + c_\beta V$. Substitution proves (5.2), even with coefficient $1/\beta$ instead of $a_{\beta,\delta}$ in this case.

For $V > 1 - \delta$, apply the entropy variational inequality to $\beta \log(\ell q)$, obtaining

$$D(p\|u_\ell) \leq c_\beta D(p\|q) + \beta^{-1} \log R.$$

Since $V/(1-\delta) > 1$ and $\log(R/V) \geq \log R \geq 0$, this is bounded by (5.2). The uniform case and zeros follow by continuity. $\square$

Proposition 5.3 (Explicit improved clock). *For $1 \leq B \leq n$, put*

$$J = \left\lceil \frac{\log(n/B)}{L} \right\rceil, \quad Z = LJ, \quad \delta = \frac{1}{Z+4}, \quad \beta = \beta_c - \frac{1}{Z+4}.$$

Use the constants of Lemma 5.2, and define

$$\begin{aligned} D_0 &= \frac{a_{\beta,\delta}}{L}, \\ F_0 &= c_\beta + \frac{b_{\beta,\delta}}{L} + D_0 \left[\log^+(C_\beta L) + \log(100 + 12Z) + \frac{1}{e} \right], \\ R_*(B) &= \left\lceil 100 \log^+(dL) \right\rceil + \sum_{j=1}^J \left\lceil L(F_0 + D_0 j L) \right\rceil. \end{aligned} \quad (5.3)$$

Then $\mathcal{E}(q_{R_*(B)}) \leq B$. For fixed $\eta > 0$ and $B = \eta n e^{-s} \in [1, n]$, as $s \rightarrow \infty$,

$$R_*(B) = As^2 + O_\eta(s \log(s+2) + \log(d+2)), \quad A = \frac{1}{2 \log(3/2)}. \quad (5.4)$$

Proof. Summing (5.2), using the endpoint moment and log-sum inequality (also over conditioning atoms), gives

$$E \leq c_\beta \mathcal{C} + a_{\beta,\delta} S \log(C_\beta n/S) + b_{\beta,\delta} S.$$

Let $x = E/\Delta \geq 1$. For $v = LS/\Delta \in (0, 1]$, use $v \log(1/v) \leq 1/e$ to obtain

$$S \log(C_\beta n/S) \leq \frac{\Delta}{L} [\log^+(C_\beta L n/\Delta) + 1/e].$$

Consequently, with $r = \log^+(n/E)$,

$$x \leq c_\beta + b_{\beta,\delta}/L + D_0 [\log^+(C_\beta L) + r + \log x + 1/e].$$

When $E \geq ne^{-Z}$, the coarse profile (4.3) bounds $\log x \leq \log(100 + 12Z)$. Thus $x \leq F_0 + D_0 r$. While entropy is in the j th dyadic band, this is at most $F_0 + D_0 jL$, so the corresponding summand of (5.3) suffices to halve the previous threshold. The initial term reduces entropy from at most ndL to n using (4.3). This proves the finite assertion, including $J = 0$.

The chosen β is bounded away from zero, and

$$D_0 = \frac{1}{\beta_c L} + O((Z+4)^{-1}), \quad C_\beta = O(Z+4), \quad F_0 = O(\log(Z+2)).$$

Summing the ceilings gives $R_*(B) = \frac{1}{2} D_0 Z^2 + O(Z \log(Z+2) + \log(d+2))$. Since $\beta_c L = \log(3/2)$ and $Z = s + \log(1/\eta) + O(1)$, this is (5.4). All constants are uniform as the moment order approaches its critical endpoint at the stated rate. $\square$

6 Ordered-tuple multiplicity amplification

This section is the algebraic bridge from “every small ordered collection of cards is smooth” to “the full permutation is smooth.” A reader unfamiliar with representation theory may read Lemma 6.2 as the bridge theorem and then inspect its proof for the bookkeeping that produces the binomial factors.

The regular representation of S_n decomposes functions on S_n into orthogonal matrix-valued Fourier modes indexed by partitions $\lambda \vdash n$. Tracking an ordered h -tuple of distinct labels gives another representation: S_n acts by permuting the set Ω_h . The key fact is that a low-level Fourier mode occurs with large, explicitly computable multiplicity inside this ordered-tuple action. This is why information about only h cards can control a much larger part of the full-deck distribution.

A *standard Young tableau* of a (possibly skew) Young diagram is a filling of its boxes by consecutive integers, increasing from left to right in every row and from top to bottom in every column. The branching rule identifies the multiplicity below with the number of such tableaux of an appropriate skew shape.

All statements in this section concern arbitrary probability laws on S_n , not just the Thorp walk. For a partition $\lambda \vdash n$, let V_λ be its irreducible representation and $f^\lambda = \dim V_\lambda$. Its level is

$$\ell(\lambda) = \min(n - \lambda_1, n - \lambda'_1).$$

Let E_k be the orthogonal projection in $L^2(U)$ onto the sum of all isotypic components of level k , and let $E_{>K}$ include every level larger than K . Level zero consists of the constant and sign characters.

For convolution by a probability density f , write T_f for the operator and $\|T_f\|_k$ for its norm on level k .

For an orthogonal realization ρ_λ , put

$$\widehat{f}(\lambda) = \mathbb{E}_{g \sim fU} \rho_\lambda(g).$$

Plancherel and the induced action on ordered tuples give

$$\|E_k f\|_2^2 = \sum_{\ell(\lambda)=k} f^\lambda \left\| \widehat{f}(\lambda) \right\|_{\text{HS}}^2, \quad (6.1)$$

$$\left\| P_h^f \right\|_{\text{HS}}^2 = \sum_{\lambda \vdash n} m_h(\lambda) \left\| \widehat{f}(\lambda) \right\|_{\text{HS}}^2. \quad (6.2)$$

Here P_h^f is the stochastic tuple transition matrix and $m_h(\lambda)$ its representation multiplicity. Hilbert–Schmidt norm in (6.2) is the ordinary matrix norm $\sum_{x,y} (P_h^f(x,y))^2$; no additional N_h factor is present.

Lemma 6.1 (Exact multiplicity and dimension ratio). *Let $1 \leq k \leq h \leq n - k$ and $\lambda = (n - k, \mu)$ with $\mu \vdash k$. Then*

$$m_h(\lambda) = \binom{h}{k} f^\mu, \quad f^\lambda \leq \binom{n}{k} f^\mu. \quad (6.3)$$

Proof. The ordered-tuple module is $\text{Ind}_{S_{n-h}}^{S_n} \mathbf{1}$. By reciprocity and iterated Young branching, $m_h(\lambda)$ is the number of standard tableaux of the skew shape $\lambda/(n-h)$; the branching rule is recalled in [5, Section 2.2]. Because $n - h \geq k \geq \mu_1$, this skew shape is a separated row of length $h - k$ and the lower diagram μ . Choose the k entries placed in μ and fill it in f^μ ways; the other row is forced. This proves the equality. In a full standard tableau of shape λ , choosing the k lower entries and a tableau of shape μ determines the first row, but some choices may violate column inequalities. This gives the upper bound. $\square$

Lemma 6.2 (Unsigned and signed marked bounds). *Let f be a probability density and $\mathcal{B}_h = \left\| P_h^f \right\|_{\text{HS}}^2$. Suppose $1 \leq k < (n - 1)/2$ and $k \leq h \leq n - k$. Then*

$$\|T_f\|_k^2 \leq \frac{\mathcal{B}_h}{\binom{h}{k}}, \quad (6.4)$$

$$\|E_k f\|_2^2 \leq 2\mathcal{B}_h \frac{\binom{n}{k}}{\binom{h}{k}}, \quad (6.5)$$

$$\|E_k(f * f * f)\|_2^2 \leq 2\mathcal{B}_h^3 \frac{\binom{n}{k}}{\binom{h}{k}^3}. \quad (6.6)$$

Proof. For the row wing $\lambda_1 = n - k$, (6.2) and (6.3) imply

$$\left\| \widehat{f}(\lambda) \right\|_{\text{op}}^2 \leq \left\| \widehat{f}(\lambda) \right\|_{\text{HS}}^2 \leq \frac{\mathcal{B}_h}{\binom{h}{k} f^\mu} \leq \frac{\mathcal{B}_h}{\binom{h}{k}}.$$

Also $f^\lambda / m_h(\lambda) \leq \binom{n}{k} / \binom{h}{k}$, so summing the nonnegative terms in (6.2) proves the energy bound for that wing without any factor counting partitions.

For the column wing, use the sign-twisted tuple module. Its matrix has entries

$$\sum_g f(g)U(g)\text{sgn}(g)\mathbf{1}_{gx=y},$$

whose absolute values are at most $P_h^f(x, y)$. Its squared Hilbert–Schmidt norm is therefore at most $\mathcal{B}_h$. The multiplicity of λ in this module is the unsigned multiplicity of λ' . This proves the same bounds for the column wing. The two wings are disjoint in the stated range, giving the factor two in (6.5). Finally matrix multiplication and $\|ABC\|_{\text{HS}} \leq \|A\|_{\text{HS}} \|B\|_{\text{op}} \|C\|_{\text{op}}$ give (6.6). No matrix normality or eigenvalue estimate is used. $\square$

7 Truncation preserves the marked budget

The purpose of this section is elementary but important. Relative entropy controls an average of $F \log F$, whereas the high-level representation bound needs an L^2 norm. A tiny set on which F is enormous can have modest entropy but terrible L^2 norm. We isolate that rare set exactly.

Choose a threshold M . The mass of the set $\{F > M\}$ is at most η once M is chosen from the entropy bound. Conditioning on the complementary event produces a probability density f with $f \leq F/p$ and controlled L^2 norm; the removed mass becomes another probability density b . Thus $F = pf + (1 - p)b$ is an identity of probability laws. Nothing is discarded from the actual shuffle, and the exceptional probability $1 - p$ is tracked all the way to the final total-variation estimate.

Lemma 7.1 (Entropy truncation with pointwise domination). *Let F be a probability density with $\mathcal{E}(F) \leq B$, and let $0 < \eta < 1$. Set $\log M = (B + 1/e)/\eta$. There is an exact decomposition*

$$F = pf + (1 - p)b, \quad p \geq 1 - \eta, \quad (7.1)$$

into probability densities, with

$$f \leq F/p, \quad \log \|f\|_2 \leq \frac{B + 1/e}{2\eta} + \frac{1}{2} \log \frac{1}{1 - \eta}. \quad (7.2)$$

If $N_h P_h^F(x, y) \leq e^{\alpha h^2}$, then

$$\|P_h^f\|_{\text{HS}}^2 \leq p^{-1} e^{\alpha h^2}. \quad (7.3)$$

Proof. Let $a = \mathbb{E}_U[F \mathbf{1}_{F > M}]$. Since $x \log x \geq -1/e$, $a \log M \leq \mathcal{E}(F) + 1/e \leq B + 1/e$, so $a \leq \eta$. Set $p = 1 - a$, $f = F \mathbf{1}_{F \leq M}/p$, and normalize the tail as b ; when $a = 0$, the choice of b is immaterial. Then $\|f\|_2^2 \leq M/p$, proving (7.2). Domination passes to every tuple transition entry. Because P_h^f is stochastic,

$$\sum_{x,y} P_h^f(x, y)^2 \leq \max_{x,y} P_h^f(x, y) \sum_{x,y} P_h^f(x, y) \leq \frac{e^{\alpha h^2}}{p N_h} N_h.$$

This proves (7.3). $\square$

The last inequality is why entropy preparation and marked smoothing can be performed by the *same* actual block. It is not an assertion that one may overlap independent factors without paying their time.

Proposition 7.2 (Optimized marked order). *Suppose $t \geq 2m$, $F = q_t$, and f is as in Lemma 7.1. Let α be (4.1) and assume*

$$\alpha n \geq 16, \quad 1 \leq K < n/10, \quad \alpha K \leq 1/16. \quad (7.4)$$

Then, for $1 \leq k \leq K$,

$$\|T_f\|_k^2 \leq p^{-1} (2e\sqrt{\alpha k})^k, \quad (7.5)$$

$$\|E_k f\|_2^2 \leq p^{-1} (4en\sqrt{\alpha/k})^k, \quad (7.6)$$

$$\|E_k f^{*3}\|_2^2 \leq p^{-3} (16e^3 n \alpha^{3/2} \sqrt{k})^k. \quad (7.7)$$

Proof. For this level choose $h = \lfloor \sqrt{k/\alpha} \rfloor$ and put $r = \sqrt{k/\alpha}$. Then $r \geq 4k$, $h \geq r/2 \geq k$, and $h - k + 1 \geq r - k \geq r/2$. Also $h \leq \sqrt{n/\alpha} \leq n/4 \leq n - k$ and $\alpha h \leq \sqrt{\alpha k} \leq 1/4 \leq m - 1$. Corollary 4.1 and truncation give $\mathcal{B}_h \leq p^{-1} e^{\alpha h^2} \leq p^{-1} e^k$. Use

$$\binom{h}{k} \geq (h/k)^k, \quad \frac{\binom{n}{k}}{\binom{h}{k}} = \prod_{j=0}^{k-1} \frac{n-j}{h-j} \leq \left(\frac{n}{h-k+1} \right)^k.$$

Inserting these in Lemma 6.2 gives (7.5) and (7.6), with $2 \leq 2^k$ absorbing the two wings. Their product gives (7.7). The marked order is allowed to depend on k ; all estimates concern the same density f . $\square$

8 Two regular blocks and one actual smoothing block

We now assemble the proof using actual consecutive shuffle blocks. It is useful to keep the roles distinct:

- a *long block* of length t sweeps is run long enough both to lower its entropy and to smooth every marked-card order needed later;
- from each long block we analyze only its high-probability regular component f , while paying at most η for the exceptional component;
- a *short block* of length u sweeps is kept as its actual density $G = q_u$. It is not truncated and need not satisfy an entropy target. Its only job is to provide one more low-level contraction.

Under our convolution convention the regular component to estimate is $g = G * f * f$. Because the original blocks are powers of the same sweep kernel, the corresponding actual law is exactly $q_u * q_t * q_t = q_{2t+u}$. The proof below never reorders G, f, f after the two truncations.

The parameters in Theorem 8.1 have the following roles. The number H is a logarithmic L^2 budget for f ; $K = \lfloor H \rfloor$ is the Fourier level at which we switch from ordered-tuple control to the high-level dimension bound. The quantities α and β_0 measure the remaining marked-card interaction in the long and short blocks, respectively. Finally $\rho = 16e^3 n \alpha \sqrt{\beta_0 K}$ is the common ratio of the low-level geometric series. The theorem simply requires $\rho < 1$ and makes every error term explicit.

The high-level dimension estimate is

$$\|T_f\|_{>K} \leq \|f\|_2 (eK/n)^{K/2}, \quad 1 \leq K < n/10. \quad (8.1)$$

This is the external estimate (2.3); see [5, Theorem 4.4 and Lemma 4.5]. Every probability convolution contracts every central isotypic level.

Theorem 8.1 (Asymmetric finite criterion). *Choose $0 < \eta < 1/3$, $H \geq 2$, $B = \eta H \in [1, n]$, and*

$$H \geq \frac{1}{e\eta} + \log \frac{1}{1-\eta}, \quad K = \lfloor H \rfloor < n/10.$$

Let $m_a, m_b \geq 2$ be integers, $\alpha = \alpha_{d, m_a}$, $\beta_0 = \alpha_{d, m_b}$, and

$$t = \max\{R_*(B), 2m_a\}, \quad u = 2m_b.$$

Suppose $\alpha n, \beta_0 n \geq 16$, $\alpha K, \beta_0 K \leq 1/16$, and

$$\rho = 16e^3 n \alpha \sqrt{\beta_0 K} < 1.$$

Then

$$\|q_{2t+u}U - U\|_{\text{TV}} \leq 2\eta + \frac{1}{2} \left[(1-\eta)^{-2} \frac{\rho}{1-\rho} + e^{4H} (eK/n)^K \right]^{1/2}. \quad (8.2)$$

The physical time is exactly $d(2t+u)$ ordinary Thorp shuffles.

Proof. Let $F = q_t$. The improved clock gives $\mathcal{E}(F) \leq B$, and the marked kernel bound holds simultaneously at every required order. Lemma 7.1 gives $F = pf + (1-p)b$, $p \geq 1-\eta$, with $\log \|f\|_2 \leq H$. Two independent copies have regular component $f * f$ with weight p^2 ; the other components have mass at most 2η . Put $G = q_u$, without truncating or imposing an entropy requirement on it.

The proof of Proposition 7.2, applied to G with $p = 1$ and parameter β_0 , gives

$$\|T_G\|_k^2 \leq (2e\sqrt{\beta_0 k})^k, \quad 1 \leq k \leq K.$$

This uses only the marked kernel estimate, not an entropy hypothesis. For the regular component $g = G * f * f$, the same proposition gives

$$\begin{aligned} \|E_k g\|_2^2 &\leq \|T_G\|_k^2 \|T_f\|_k^2 \|E_k f\|_2^2 \\ &\leq p^{-2} (16e^3 n \alpha \sqrt{\beta_0 k})^k. \end{aligned}$$

Summation over $k \leq K$ gives the first term in the square brackets of (8.2). For high levels, probability contraction and (8.1) yield

$$\|E_{>K} g\|_2^2 \leq \|E_{>K}(f * f)\|_2^2 \leq \|f\|_2^4 (eK/n)^K \leq e^{4H} (eK/n)^K.$$

The actual factor $G = q_u$ has sign coefficient zero, because it includes at least one fair-switch sweep. Therefore g has only its constant coefficient on level zero. No extra parity-removal sweep is needed. Orthogonality, $\text{TV} \leq \frac{1}{2}L^2$, and the exact mixture prove the bound. The actual law is $q_u * q_t * q_t = q_{2t+u}$. No factors were reordered after regularization, and no artificial kernel replaced an actual block. $\square$

9 Choice of parameters and optimization of the coefficient

We first explain the scale before choosing the constants. Put $H \approx ne^{-s}$. Reaching entropy of this order costs approximately As^2 sweeps, where $A = 1/(2\log(3/2))$. On the other hand, the marked-card bound contains an exponential factor of the form $\exp\{cd/m\}$, so making it compatible with the cutoff $H \approx ne^{-s}$ requires a smoothing depth m of order d/s . Hence the long-block duration is essentially

$$\max\{As^2, cd/s\}.$$

Balancing the two mechanisms forces $s \asymp d^{1/3}$ and a block length $\asymp d^{2/3}$ sweeps. The remaining parameter $z > 2$ controls how the smoothing burden is divided between the two long blocks and the shorter third block. The calculation below chooses the value of z that minimizes the leading constant.

Fix $\eta = \varepsilon/8$. For sufficiently large d , set

$$\begin{aligned} v &= (z_* Ld/A)^{1/3}, & s &= v + 4\log(d+2) + 100, \\ H &= ne^{-s}, & B &= \eta H, \quad K = \lfloor H \rfloor, \\ m_a &= 2 + \left\lceil \frac{z_* L(d+1)}{2v} \right\rceil, & m_b &= 2 + \left\lceil \frac{z_* L(d+1)}{2(z_* - 2)v} \right\rceil. \end{aligned} \tag{9.1}$$

Let t, u be as in Theorem 8.1. Because $s = o(d)$, all entropy and norm-budget domains eventually hold. The two marked parameters satisfy

$$\alpha \leq \frac{8Ld}{n} e^{v/z_*}, \quad \beta_0 \leq \frac{8Ld}{n} e^{(z_* - 2)v/z_*}.$$

Both are at least $8Ld/n$, and both multiplied by K tend to zero. The low-level ratio is bounded by

$$\begin{aligned} \rho &\leq 16e^3 (8Ld)^{3/2} \exp \left\{ \frac{v}{z_*} + \frac{(z_* - 2)v}{2z_*} - \frac{s}{2} \right\} \\ &= 16e^3 (8Ld)^{3/2} e^{(v-s)/2} = O(d^{-1/2}). \end{aligned}$$

For the high-level term, $H/2 \leq K \leq H$ eventually, so

$$4H + K \log(eK/n) \leq H(4.5 - s/2) \longrightarrow -\infty.$$

The right side of (8.2) is thus eventually less than ε .

The times satisfy

$$\begin{aligned} R_*(B) &= Av^2 + O_\varepsilon(v \log(d+2) + \log^2(d+2)), \\ 2m_a &= \frac{z_* Ld}{v} + O(1) = Av^2 + O(1), \\ u = 2m_b &= \frac{z_* Ld}{(z_* - 2)v} + O(1) = \frac{Av^2}{z_* - 2} + O(1). \end{aligned}$$

Taking the genuine maximum for t and adding both long blocks and the short block gives

$$d(2t + u) = \left(2 + \frac{1}{z_* - 2}\right) A^{1/3} (z_* L)^{2/3} d^{5/3} + O_\varepsilon(d^{4/3} \log(d+2)).$$

This proves Theorem 1.1. Group translation gives every initial deck. Finitely many dimensions are absorbed into the error-term constant: identity has positive sweep probability and hypercube-edge transpositions belong to sweep support and generate S_n .

Why this particular algebraic number? Within this two-equal-regular-block, one-actual-block family, any fixed $z > 2$ yields the coefficient

$$A^{1/3}L^{2/3}F(z), \quad F(z) = z^{2/3} \left(2 + \frac{1}{z-2} \right).$$

One has

$$F'(z) = \frac{4z^2 - 17z + 12}{3z^{1/3}(z-2)^2}.$$

Since F diverges at both ends of $(2, \infty)$, its unique minimizer is $z_* = (17 + \sqrt{97})/8$. This is optimization only within this stated family, not an optimality assertion for Thorp mixing. The actual shorter block has asymptotic length $1/(z_* - 2) = 0.7374048168\dots$ times that of each regular block.

10 Pitfalls, limitations, and the remaining gap

Two shortcuts that might appear to improve the argument are invalid and are worth recording. First, unordered subsets cannot replace ordered tuples in Section 6; the exact multiplicity $\binom{h}{k} f^\mu$ is an ordered-tuple statement. Second, a uniform two-sweep pointwise estimate of the form $n!q_2(\sigma) \leq e^{Cn}$ with an absolute constant C is false. To see the latter, choose a matching of $n/8$ cube edges and activate exactly those switches in both sweeps. Greedy counting gives at least

$$\frac{(nd/4)^{n/8}}{(n/8)!} \geq (2d)^{n/8}$$

such choices. Each selected edge is traversed twice and returns its two cards, so the resulting permutation is the identity. Since two sweeps have n^n equiprobable switch configurations,

$$\log(n!q_2(\text{id})) \geq \frac{n}{8} \log(2d) - n.$$

This obstruction is only pointwise; it neither rules out an $O(n)$ entropy bound after two sweeps nor yields a $d^{5/3}$ lower bound.

The method still balances a quadratic entropy-preparation cost in a parameter s against a marked-card smoothing cost of order d/s sweeps. Their balance produces the power $d^{2/3}$ in sweep count, hence $d^{5/3}$ ordinary Thorp shuffles. Improving the exponent therefore appears to require a genuinely stronger entropy clock, a stronger marked-card smoothing estimate, or a new finishing mechanism. No cutoff statement, limiting profile, or matching lower bound is proved here.

Detailed analytic inputs

For completeness, we now prove the shuffle-specific analytic statements used in the main argument. The only external negative-dependence input is (E1) in Section 2.2; all remaining estimates below are derived explicitly.

A Collision graphs and inclusion probabilities

A sweep is a network of d matching layers. Once all switch bits in that network are fixed, every input vertex follows one deterministic path through the network. Pulling each physical matching edge backwards along those paths creates a graph on the *input* vertices. Two marked input cards collide in the sweep exactly when their two input vertices form an edge of this pulled- back graph. The next lemmas formalize this picture.

Let Ω_k be the ordered distinct k -tuples of vertices, let $N_k = (n)_k = n(n-1) \cdots (n-k+1)$, and give Ω_k uniform probability π_k . Let P_k be the transition matrix of one sweep on these tuples.

Lemma A.1 (Exact labelled routing weight). *For $x, y \in \Omega_k$, the one-sweep route is either infeasible or its collision count $C(x, y)$ is determined by the endpoints and*

$$P_k(x, y) = 2^{-kd+C(x,y)} = n^{-k} 2^{C(x,y)}. \quad (\text{A.1})$$

A collision is a matching edge containing two marked cards immediately before its layer.

Proof. Before layer i , a card with initial vertex x_a and terminal vertex y_a must be at $(y_{a,<i}, x_{a,\geq i})$. Thus every marked path is prescribed. An inconsistent prescription is infeasible. Otherwise it prescribes one fair switch for each visited edge. If c_i edges are doubly occupied at layer i , there are $k - c_i$ visited edges at that layer. Consequently the number of independent prescribed switches is $kd - \sum_i c_i$. Unvisited switches impose no condition. $\square$

Fix a realization ω of all switches in a sweep. Let T_i map input vertices to their positions immediately before layer i . Pull that layer's matching back by T_i^{-1} and take the union over i ; denote the resulting graph by G_ω .

Lemma A.2 (Uniform collision-graph geometry). *For every ω , the graph G_ω is simple and d -regular. The number of collisions of a marked input set A is exactly $e_{G_\omega}(A)$. Every nonempty vertex set B satisfies*

$$e_{G_\omega}(B) \leq \frac{|B|}{2} \log_2 |B|. \quad (\text{A.2})$$

The same statements hold in the reverse coordinate order.

Proof. Every pulled-back matching has degree one. Endpoints of a layer- i edge have highest differing initial coordinate i , because previous layers changed only lower coordinates. Thus edges from distinct layers do not repeat. The collision identity is the definition of pullback.

Fix the suffix above coordinate i . Let a, b be the numbers of vertices of B having that suffix and having bit i equal to zero or one. Earlier layers do not change these counts, so at most $\min(a, b)$ of the layer- i matching edges connect two such vertices. For binary entropy h_2 ,

$$\min(a, b) \leq \frac{a+b}{2} h_2\left(\frac{a}{a+b}\right),$$

with the empty-node contribution defined as zero. This follows from $h_2(p) \geq 2 \min(p, 1-p)$. Summing over the suffix tree telescopes to $|B| \log_2 |B|/2$, since its terminal nonempty leaves have size one. Reverse the suffix tree for the reverse-order sweep. $\square$

Lemma A.3 (One-sweep inclusion bound). *Start from any deterministic k -set, and let A be its occupied set after one sweep. Then, for every $B \subseteq V$,*

$$\mathbb{P}(B \subseteq A) \leq (k/n)^{|B|}. \quad (\text{A.3})$$

This also holds for a reverse-order sweep.

Proof. The generating polynomial of a deterministic set is a monomial and is stable: it is nonzero when all variables have positive imaginary parts. A measure with stable multiaffine generating polynomial and nonnegative coefficients is strongly Rayleigh. Averaging a measure with its image under a transposition is partial symmetrization. By [3, Theorem 4.20], it preserves this property, and [3, Theorem 4.9] supplies negative association. Apply these statements to the independent edge swaps in succession.

One labelled card is exactly uniform after a sweep: every coordinate becomes a fresh fair bit at its sole update. Thus every vertex has occupation probability k/n . Negative association, applied repeatedly to distinct occupation indicators, proves (A.3). If $|B| > k$ the left side is zero. The argument is independent of the coordinate order. $\square$

B A sharper unrestricted collision moment

The preceding section reduced marked-card dependence to the random variable $e_G(A)$: the number of edges induced by the occupied marked set A in a fixed collision graph G . We now bound an exponential moment of this count. The proof expands the product over edges, groups selected edges into connected components, and overcounts each component by an embedded spanning tree. This is the step that turns cube geometry plus negative dependence into a usable analytic bound.

The next result uses only the inclusion bound, not the full strong-Rayleigh property.

Theorem B.1 (Connected edge-set moment). *Let G be a simple graph on n vertices, of maximum degree at most d , satisfying (A.2). Let A be a random set of exactly $k \geq 2$ vertices satisfying (A.3). For $\lambda > 0$, define*

$$a = 4d \frac{k}{n} (1 - 2^{-\lambda})(2k)^{\lambda/2}. \quad (\text{B.1})$$

If $a < 1$, then

$$\mathbb{E} 2^{\lambda e_G(A)} \leq \exp\left(\frac{ka}{1-a}\right). \quad (\text{B.2})$$

Proof. Put $t = 2^\lambda - 1 > 0$ and $\theta = k/n$. Expand over subsets of edges:

$$2^{\lambda e_G(A)} = \prod_{e \in E(G)} (1 + t \mathbf{1}_{\{e \subseteq A\}}) = \sum_{F \subseteq E(G)} t^{|F|} \mathbf{1}_{\{V(F) \subseteq A\}}, \quad (\text{B.3})$$

where $V(F)$ is the set of endpoints of the chosen edges. Every nonzero term has $|V(F)| \leq k$. For $B \subseteq V$ with $2 \leq |B| \leq k$, define

$$w(B) = \sum_{\substack{F \subseteq E(G[B]) \\ (B, F) \text{ connected}}} t^{|F|}.$$

Decomposing F into its connected components, using the inclusion bound on their disjoint union, and then allowing all collections, overlaps, and repetitions in an exponential series yields

$$\mathbb{E} 2^{\lambda e_G(A)} \leq \exp \left(\sum_{2 \leq |B| \leq k} \theta^{|B|} w(B) \right). \quad (\text{B.4})$$

More explicitly, before the last enlargement the sum is over unordered pairwise disjoint collections of connected vertex sets. Its summand is the product of their weights $\theta^{|B|} w(B)$. All terms are nonnegative, so the enlargement is an upper bound. It asserts no independence of components.

If $|B| = v$ and (B, F) is connected, F contains a spanning tree. Summing over all possible spanning trees can only overcount, so

$$w(B) \leq \sum_{T \text{ spanning tree of } G[B]} t^{v-1} (1+t)^{e_G(B)-(v-1)}. \quad (\text{B.5})$$

The total number of pairs (B, T) with $|B| = v$ is at most $n(4d)^{v-1}$. To see this, encode a rooted plane version of T by its root, one of at most 4^{v-1} rooted plane tree shapes, and a neighbor index, with at most d choices, for each tree edge. For example, choose the smallest-labelled root and a fixed ordering of children to assign one encoding to every pair. Such an encoding determines the embedded tree, so it is a valid overcount of *tree pairs*, not merely of vertex sets.

By (A.2), $(1+t)^{e_G(B)} \leq v^{\lambda v/2}$. Also, for $v \geq 2$,

$$v^{v/(v-1)} \leq 2v \leq 2k, \quad v^{\lambda v/2} \leq (2k)^{\lambda(v-1)/2}.$$

Here $v \leq 2^{v-1}$ proves the first inequality. Therefore the exponent in (B.4) is at most

$$\begin{aligned} \sum_{v=2}^k n(4d)^{v-1} \theta^v \left(\frac{t}{1+t} \right)^{v-1} v^{\lambda v/2} &\leq k \sum_{v=2}^k \left[4d\theta(1-2^{-\lambda})(2k)^{\lambda/2} \right]^{v-1} \\ &\leq \frac{ka}{1-a}. \end{aligned}$$

This proves (B.2). □

Remark B.2 (The gain and the independence condition). The preceding proof retains one factor $1 - 2^{-\lambda}$ for every edge of a spanning tree. The old vertex-component bound omitted these factors, giving $a_{\text{old}} = 4d(k/n)(2k)^{\lambda/2}$. Thus the new theorem is stronger and remains useful as $\lambda \downarrow 0$.

In the shuffle application, A is produced by the first sweep, and G is the collision graph of an independent second sweep. One may fix the entire second network first. Its independence of A preserves (A.3), and the theorem is uniform over every network. There is no restriction to a low-collision event.

C From collision moments to a pointwise heat kernel

The collision moment is an average bound. The final representation argument needs a pointwise bound on the marked-card transition kernel. We obtain it by viewing two sweeps as an operator $K = P_k^2$, proving matching row and column L^p estimates, and then interpolating from L^1 to L^∞ in m steps. Reverse-coordinate sweeps supply the column estimate; no reversibility of a complete sweep is assumed.

Theorem C.1 (Finite-parameter marked-card smoothing). *Let $m \geq 2$ be an integer and $2 \leq k \leq n$. Put*

$$a_{d,k,m} = 4d \frac{k}{n} (1 - 2^{-1/(m-1)}) (2k)^{1/[2(m-1)]}. \quad (\text{C.1})$$

If $a_{d,k,m} < 1$, then, for all ordered distinct endpoints x, y ,

$$N_k P_k^{2m}(x, y) \leq \exp \left(\frac{(m-1)ka_{d,k,m}}{1 - a_{d,k,m}} \right). \quad (\text{C.2})$$

In particular the sufficient condition

$$8(\log 2)d \frac{k}{n} (2k)^{1/[2(m-1)]} \leq 1 \quad (\text{C.3})$$

implies $N_k P_k^{2m}(x, y) \leq e^k$. Both conclusions hold for inverse noise. For $k = 1$, the normalized kernel is already one after a single sweep.

Proof. Set $p = m/(m-1)$, $p' = m$, $\lambda = p-1$, $N = N_k$ and $K = P_k^2$. Jensen's inequality over the first-sweep transition law gives

$$\begin{aligned} \frac{1}{N} \sum_y (NK(x, y))^p &\leq \sum_z P_k(x, z) \frac{1}{N} \sum_y (NP_k(z, y))^p \\ &= \sum_z P_k(x, z) \sum_y P_k(z, y) (NP_k(z, y))^{p-1} \\ &\leq \mathbb{E}_x 2^{(p-1)C_2} \leq \exp \left(\frac{ka_{d,k,m}}{1 - a_{d,k,m}} \right), \end{aligned} \quad (\text{C.4})$$

where C_2 is the second sweep's collision count. The penultimate inequality uses Lemma A.1 and $Nn^{-k} \leq 1$. The last uses Theorem B.1, with the independent future graph as just explained.

Every coordinate-layer operator is self-adjoint, so P_k^* is the actual reverse-order sweep. Repeating (C.4) for $(P_k^*)^2$ gives the identical column $L^p(\pi_k)$ bound. With

$$M = \exp \left(\frac{ka_{d,k,m}}{p(1 - a_{d,k,m})} \right),$$

Minkowski and Hölder therefore give, respectively, $\|K\|_{L^1 \rightarrow L^p} \leq M$ and $\|K\|_{L^{p'} \rightarrow L^\infty} \leq M$. Finite-dimensional Riesz–Thorin interpolation gives

$$\|K\|_{L^r \rightarrow L^s} \leq M, \quad \frac{1}{s} = \frac{1}{r} - \frac{1}{m}, \quad 1 \leq r \leq m.$$

Iterate at $r_j = m/(m-j)$, $j = 0, \dots, m$, with $r_0 = 1$ and $r_m = \infty$. All m applications lie in the permitted range. Hence

$$\|K^m\|_{L^1 \rightarrow L^\infty} \leq M^m = \exp \left(\frac{(m-1)ka_{d,k,m}}{1 - a_{d,k,m}} \right).$$

For a nonnegative matrix on uniform probability space, $\|K^m\|_{L^1 \rightarrow L^\infty} = N \max_{x,y} K^m(x, y)$; a normalized point mass attains the maximum. This proves (C.2).

For the constant-base assertion, write $c_* = 4Ld(k/n)(2k)^{1/[2(m-1)]}$. The elementary bound $(m-1)(1-2^{-1/(m-1)}) \leq L$ gives $a_{d,k,m} \leq c_*/(m-1)$. If $c_* \leq 1/2$, then

$$\frac{(m-1)a_{d,k,m}}{1-a_{d,k,m}} \leq \frac{c_*}{1-c_*/(m-1)} \leq 1.$$

This is precisely (C.3). Reverse sweeps prove the inverse claim. No reversibility of the sweep kernel is assumed. $\square$

D Endpoint geometry and the two entropy losses

This is the geometric heart of the entropy argument. The idea is to reveal the endpoints of a fresh sweep from the largest input index downward. Even after higher endpoints are known, the next endpoint is uniform on an explicit candidate set. Inside that set we construct a symmetric random partner. When we forget the detailed endpoint information, the partner's preimage is exactly uniform among the earlier input positions. This produces two complementary ways to certify that the sweep erased information.

This section supplies the shuffle-specific information inequalities needed for the improved entropy clock. They apply to an arbitrary input law, not only to a law started from the identity.

Let X be one independent sweep as a position permutation, let μ be a random arrangement mapping labels to positions, and put $Y = X \circ \mu$. The entropy loss is

$$\Delta = \mathcal{E}(\mu) - \mathcal{E}(Y) = I(Y; X), \quad (\text{D.1})$$

because the Shannon entropy of Y given X equals that of μ . Index positions by the integers $0, \dots, n-1$, using their bit expansions. Set

$$\mathcal{H}_j = \sigma(X(i) : i > j), \quad A_j = X(j),$$

and let $s(j)$ be the number of one-bits of j .

Lemma D.1 (Candidates and a symmetric partner). *Given $\mathcal{H}_j$, A_j is uniform on a measurable candidate set C_j of size $2^{s(j)}$, containing no endpoint of a higher input. For $j \geq 1$ there is a measurable symmetric stochastic matrix Q_j on C_j , with zero diagonal, such that a draw B_j from $Q_j(A_j, \cdot)$ using independent auxiliary randomness satisfies*

$$K_j := X^{-1}(B_j) < j, \quad \mathbb{P}(K_j = k) = 1/j \quad (0 \leq k < j). \quad (\text{D.2})$$

The displayed probabilities average the fresh sweep. They are not conditional probabilities given $\mathcal{H}_j$.

Proof. Induct through a cube of size $2a$. Its first $d-1$ layers are independent sweeps on the lower and upper halves, and its last layer matches equal lower coordinates across the halves. Write an upper input as $j = a + i$. The exposed higher upper endpoints reveal the corresponding smaller upper endpoints and the final switch bits at their distinct used coordinates. Each prescribed pattern of those bits has probability $2^{-(a-1-i)}$, independently of the smaller routing. It therefore does not bias that routing. The unexposed endpoint of $a + i$ is uniform on $C_i^{\text{upper}} \times \{0, 1\}$.

For a lower input, all upper endpoints have been exposed. They reveal the entire smaller upper routing and every final matching bit. The last layer is now a known bijection from lower smaller

coordinates to the remaining endpoints. It transports the candidate set from the smaller lower sweep. This proves the candidate assertion, including measurability, by induction starting with a one-vertex cube.

For lower inputs transport the smaller partner kernel by that known bijection. For $j = a + i$, from (x, b) choose $(x, 1 - b)$ with probability $(a - i)/(a + i)$. With the remaining probability $2i/(a + i)$, choose y according to the smaller $Q_i(x, \cdot)$, draw a fair independent bit c , and choose (y, c) . If $i = 0$ only the first branch is used. The resulting kernel is symmetric, stochastic, and loop-free.

In the first branch the input at the other endpoint is uniform among the a lower inputs, by independence and one-card uniformity of the smaller lower sweep. In the second branch the fair bit chooses an upper or lower input with probability $1/2$. The upper input is uniform among the i earlier upper inputs by induction; the lower input is uniform among the a lower inputs by its independent sweep. Consequently each lower input has probability

$$\frac{a - i}{(a + i)a} + \frac{i}{(a + i)a} = \frac{1}{a + i},$$

and each earlier upper input has probability $(2i/(a + i))/(2i) = 1/(a + i)$. All candidates exclude the higher inputs, and the diagonal is excluded, so $K_j < j$ holds pointwise. This establishes (D.2) and completes the induction. $\square$

For each j , let $\mathcal{G}_{j+1} = \sigma(\mu^{-1}(i) : i > j)$. On an atom of this field, the remaining label set has size $\ell = j + 1$. Write p_k for the conditional law of $\mu^{-1}(k)$ on that atom, for every $k \leq j$. All p_k in a row of the argument use this *same* conditioning field; the matrix $(p_k(a))$ on the remaining labels is doubly stochastic. Let u_ℓ be uniform on those labels. The chain rules give

$$\mathcal{E}(\mu) = \sum_{j=1}^{n-1} \mathbb{E}D(p_j \| u_{j+1}), \quad \Delta = \sum_{j=1}^{n-1} I(Y; A_j \mid \mathcal{H}_j). \quad (\text{D.3})$$

The omitted $j = 0$ terms are zero.

Define, for a distribution p on ℓ points,

$$V(p) = 1 - \frac{1}{\ell} \left(\sum_a \sqrt{p(a)} \right)^2, \quad S = \sum_{j=1}^{n-1} \mathbb{E}V(p_j).$$

Lemma D.2 (Two separate losses). *There are deterministic probability weights W_{jk} , $k \leq j$, with $W_{00} = 1$ and, for $j = a + i$ where a is the largest power of two at most j ,*

$$W_{jk} = \frac{1}{2a} \quad (k < a), \quad W_{j,a+k} = \frac{1}{2} W_{ik} \quad (k \leq i), \quad (\text{D.4})$$

such that, with $q_j = \sum_{k \leq j} W_{jk} p_k$,

$$\Delta \geq LS, \quad \Delta \geq \mathcal{C} := \sum_{j=1}^{n-1} \mathbb{E}D(p_j \| q_j). \quad (\text{D.5})$$

The two lower bounds in (D.5) are not added.

Proof. Fix j , and abbreviate $\mathcal{H}_j, \mathcal{G}_{j+1}$ to $\mathcal{H}, \mathcal{G}$. The field $\mathcal{G}$ is independent of $(A_j, \mathcal{H})$, and is recoverable from $(Y, \mathcal{H})$ because $\mu^{-1}(i) = Y^{-1}(X(i))$ for $i > j$. The conditional-information chain rule thus gives

$$I(Y; A_j \mid \mathcal{H}) = I(Y; A_j \mid \mathcal{H}, \mathcal{G}).$$

Adjoin the partner B_j of Lemma D.1 and an independent fair bit ξ . Orient (A_j, B_j) as (V_0, V_1) according to ξ . Given $\mathcal{H}$, symmetry of Q_j and uniformity of A_j make the pair exchangeable. Thus ξ is independent of $(\mathcal{H}, \mathcal{G}, V_0, V_1)$. The auxiliary draw conveys no extra information about Y given $A_j, \mathcal{H}, \mathcal{G}$. It follows that

$$\begin{aligned} I(Y; A_j \mid \mathcal{H}, \mathcal{G}) &= I(Y; A_j, B_j, \xi \mid \mathcal{H}, \mathcal{G}) \\ &= I(Y; V_0, V_1, \xi \mid \mathcal{H}, \mathcal{G}) \\ &\geq I(Y; \xi \mid \mathcal{H}, \mathcal{G}, V_0, V_1) \\ &= I(Y, \mathcal{H}, V_0, V_1; \xi \mid \mathcal{G}) \\ &\geq I((Y^{-1}(V_0), Y^{-1}(V_1)); \xi \mid \mathcal{G}). \end{aligned}$$

In the last line the endpoints and their exposing field have been forgotten. Now (D.2), independence of the sweep from μ , and the fact that K_j ranges uniformly over positions other than j show that, on an atom of $\mathcal{G}$, the two oriented label laws are R_p and R_p^T , where $p = p_j$ and

$$R_p(a, b) = \frac{p(a)}{\ell - 1} \quad (a \neq b), \quad R_p(a, a) = 0.$$

Indeed, conditional on any initial arrangement with $\mu^{-1}(j) = a$, a uniform other position carries a uniform one of the remaining labels. Writing $\bar{R}_p = (R_p + R_p^T)/2$, the last information is

$$\Phi_\ell(p) = \frac{1}{2}D(R_p \parallel \bar{R}_p) + \frac{1}{2}D(R_p^T \parallel \bar{R}_p).$$

The scalar inequality

$$a \log \frac{2a}{a+b} + b \log \frac{2b}{a+b} \geq L(\sqrt{a} - \sqrt{b})^2 \quad (\text{D.6})$$

implies $\Phi_\ell(p) \geq \ell LV(p)/(\ell - 1) \geq LV(p)$ after summing pairs. For completeness, by homogeneity set $b = 1$. If F is half the left side of (D.6) and $h(x) = (\sqrt{x} - 1)^2$, then $F''(x)/h''(x) = \sqrt{x}/(1+x)$ decreases for $x > 1$. Both functions and their first derivatives vanish at one. Successive weighted-average integrations therefore show that F/h decreases to $L/2$. Symmetry under $x \mapsto 1/x$ and continuity prove the remaining cases. Summation in (D.3) gives the first loss.

For the second loss, compare the conditional joint law of (A_j, Y) with the product of its marginals and apply the map $(a, y) \mapsto y^{-1}(a)$. Under the joint law the image has law p_j . Under the product law it has law $\sum_{k \leq j} w_{jk}(\mathcal{H}) p_k$, where

$$w_{jk}(\mathcal{H}) = \frac{\mathbb{P}(X(k) \in C_j \mid \mathcal{H})}{|C_j|}.$$

Data processing gives the corresponding relative-entropy lower bound. Convexity in the reference distribution, and independence of $\mathcal{G}$ from the sweep, replace these random weights by $W_{jk} = \mathbb{E}w_{jk}$. They are probability weights and exclude higher inputs.

The half-cube decomposition verifies (D.4). For an upper input $j = a + i$, a lower input falls in the smaller upper candidate coordinates with probability $|C_i|/a$, by the independent uniform one-card lower sweep. Dividing by $|C_j| = 2|C_i|$ gives $1/(2a)$. Upper inputs inherit half the smaller weights. Lower inputs inherit the smaller weights through the known final bijection. Sum over (D.3) to obtain the second loss. $\square$

E A dimension-free scalar interpolation inequality

The entropy proof now becomes a one-dimensional comparison between three quantities for a probability vector: relative entropy from uniformity, relative entropy from a reference law, and square-root overlap. The next lemma is deliberately dimension-free so that its constants do not deteriorate when the number of still-unrevealed labels is large.

Lemma E.1 (Elementary scalar truncation). *For $T \geq 4$ and $z \geq 0$,*

$$z \log z - z + 1 \leq (\log T + 2)(\sqrt{z} - 1)^2 + z \log^+(z/T), \quad (\text{E.1})$$

with $0 \log 0 = 0$.

Proof. For $z \leq 1$, differentiating in $\sqrt{z}$ gives $z \log z - z + 1 \leq 2(\sqrt{z} - 1)^2$. For $1 \leq z \leq T$, put $y = \sqrt{z}$. The difference between $(\log z + 2)(\sqrt{z} - 1)^2$ and $z \log z - z + 1$ is

$$3y^2 - 4y + 1 + 2(1 - 2y) \log y.$$

Its value and first derivative vanish at one, and its second derivative is $2(3y + 1)(y - 1)/y^2 \geq 0$. Increasing the coefficient from $\log z + 2$ to $\log T + 2$ preserves the inequality. For $z \geq T$, the right side minus the left side in (E.1) is

$$3y^2 - 2(\log T + 2)y + \log T + 1.$$

It is nonnegative at $y = \sqrt{T}$ by the previous case and is increasing thereafter: $3\sqrt{T} \geq \log T + 2$ for $T \geq 4$. $\square$

Lemma E.2 (Uniform scalar estimate). *Let p, q be distributions on ℓ points. Suppose*

$$\ell^{1/2} \sum_a q(a)^{3/2} \leq R, \quad 1 \leq R \leq \ell^{1/2}.$$

For $V = 1 - \ell^{-1}(\sum_a \sqrt{p(a)})^2$,

$$D(p||u_\ell) \leq 3D(p||q) + [4\log(R/V) + 8]V. \quad (\text{E.2})$$

The final term is defined as zero when $V = 0$.

Proof. Write $f = \ell p$, $A = \mathbb{E}_{u_\ell} \sqrt{f} = \sqrt{1 - V}$, and $h(z) = z \log z - z + 1$. Then

$$D(p||u_\ell) \leq \mathbb{E}_{u_\ell} A^2 h(f/A^2),$$

because the right side minus the left is $-\log A^2 - 1 + A^2 \geq 0$. If $V = 0$, p is uniform and (E.2) is immediate.

First suppose $0 < V \leq 1/2$, so $A^2 \geq 1/2$. Set $M = 8(R/V)^2$ and apply Lemma E.1 with $T = M/A^2$. The centered square integrates to V , and hence

$$D(p||u_\ell) \leq (\log M - \log A^2 + 2)V + T_1, \quad T_1 = \sum_{f(a) > M} p(a) \log(f(a)/M). \quad (\text{E.3})$$

The entropy variational inequality applied to $g = \frac{1}{3} \log^+(f/M)$ gives

$$D(p||q) \geq \frac{1}{3}T_1 - \log \mathbb{E}_q e^g.$$

This variational inequality follows directly by comparing p with the probability distribution proportional to $e^g q$ and using nonnegativity of relative entropy. By Hölder,

$$\mathbb{E}_q e^g \leq 1 + M^{-1/3} R^{2/3} p\{f > M\}^{1/3}.$$

When $f > M \geq 8$, one has $f \leq 4(\sqrt{f} - 1)^2$. Thus

$$p\{f > M\} \leq 8(1 - A) \leq 8V.$$

The extra term in the bound on $\mathbb{E}_q e^g$ is at most V by the choice of M . Consequently $T_1 \leq 3D(p\|q) + 3V$. Substituting into (E.3), and using $-\log A^2 \leq \log 2$ and $\log 16 + 5 < 8$, proves the stronger bound with $2\log(R/V)$ in place of $4\log(R/V)$.

If $V > 1/2$, apply the entropy variational inequality to $\frac{1}{2}\log(\ell q)$. Combining it with $D(p\|u_\ell) = D(p\|q) + \mathbb{E}_p \log(\ell q)$ yields

$$D(p\|u_\ell) \leq 3D(p\|q) + 2\log R \leq 3D(p\|q) + 4V\log(R/V).$$

This proves (E.2). Zeros follow by limiting conventions; if $D(p\|q) = \infty$ the conclusion is automatic. $\square$

F The improved entropy profile and clock

We can now close the feedback loop. The partner weights have a bounded summed moment below a critical exponent. Combining that moment with the two separate entropy losses yields a differential-style statement: when the current entropy is E , one sweep removes at least a controlled fraction of E . Integrating that profile over successive entropy scales gives an explicit number of sweeps needed to reach any target B .

Lemma F.1 (A subcritical endpoint moment). *For the weights of (D.4), define*

$$R_j = (j+1)^{1/2} \sum_{k \leq j} W_{jk}^{3/2}, \quad C_R = 3 + 2\sqrt{2} < 6.$$

On every conditioning atom in Section D,

$$(j+1)^{1/2} \sum_a q_j(a)^{3/2} \leq R_j, \quad 1 \leq R_j \leq (j+1)^{1/2}, \quad \sum_{j=1}^{n-1} R_j \leq C_R n. \quad (\text{F.1})$$

Proof. Doubly stochasticity gives, for each label a ,

$$\left(\sum_{k \leq j} p_k(a) W_{jk} \right)^{3/2} \leq \sum_{k \leq j} p_k(a) W_{jk}^{3/2}.$$

Sum over a to obtain the first assertion. The other bounds for an individual R_j follow from convexity and the fact that W_j is a probability row.

Let $U_j = \sum_{k \leq j} W_{jk}^{3/2}$ and $T_d = \sum_{j < 2^d} U_j$. The recursion (D.4) gives exactly

$$T_d = (1 + 2^{-3/2})T_{d-1} + 2^{-3/2}2^{(d-1)/2}, \quad T_0 = 1. \quad (\text{F.2})$$

Dividing by $2^{d/2}$ yields a scalar affine recursion with multiplier $1/\sqrt{2} + 1/4 < 1$ and additive term $1/4$. Its fixed point is $3 + 2\sqrt{2} = C_R$, which exceeds the initial value. Hence $T_d \leq C_R 2^{d/2}$, and $\sum_{j=1}^{n-1} R_j \leq n^{1/2} T_d \leq C_R n$. $\square$

Theorem F.2 (Dimension-free entropy profile). *For every nonuniform initial law μ , one independent sweep has $\Delta = \mathcal{E}(\mu) - \mathcal{E}(X \circ \mu) > 0$ and*

$$\frac{\mathcal{E}(\mu)}{\Delta} \leq 100 + 12 \log^+ \left(\frac{n}{\mathcal{E}(\mu)} \right). \quad (\text{F.3})$$

Proof. Apply Lemma E.2 on each conditioning atom and sum using (D.3), (F.1), and the log-sum inequality. Write $E = \mathcal{E}(\mu)$ and $R_\Sigma = \sum_{j=1}^{n-1} R_j$. Then

$$E \leq 3\mathcal{C} + 4S \log(R_\Sigma/S) + 8S. \quad (\text{F.4})$$

If $S = 0$, each p_j is uniform and $E = 0$. Thus $E > 0$ implies $S > 0$ and, by (D.5), $\Delta > 0$.

Put $c = 1/L \leq 3/2$ and $x = E/\Delta \geq 1$. The two separate inequalities in (D.5) give $\mathcal{C} \leq \Delta$ and $S \leq c\Delta$. For $v = S/(c\Delta) \in (0, 1]$, the elementary bound $v \log(1/v) \leq 1/e$ yields

$$\begin{aligned} S \log(C_R n/S) &= c\Delta \left[v \log \frac{C_R n}{c\Delta} + v \log(1/v) \right] \\ &\leq c\Delta \left[\log^+ \frac{C_R n}{c\Delta} + 1/e \right]. \end{aligned}$$

This avoids any monotonicity assumption on $s \log(C_R n/s)$ outside its increasing range. Dividing (F.4) by Δ gives

$$x \leq 3 + 8c + 4c/e + 4c \log^+ \frac{C_R n}{c\Delta}.$$

Since $C_R/c < 6$, $\log 6 < 2$, and $x \geq 1$, it follows that

$$x \leq 30 + 6 \log^+(n/E) + 6 \log x.$$

Concavity of the logarithm gives $\log x \leq x/12 + \log 12 - 1 \leq x/12 + 2$. Absorb $x/2$ on the right to obtain $x \leq 84 + 12 \log^+(n/E)$, and enlarge 84 to 100. $\square$

Corollary F.3 (Explicit sweep clock). *For $1 \leq B \leq n$, define*

$$\begin{aligned} J(B) &= \lceil \log_2(n/B) \rceil, \\ R(B) &= \left\lceil 100 \log^+(dL) \right\rceil + \sum_{j=1}^{J(B)} \lceil L(100 + 12jL) \rceil. \end{aligned} \quad (\text{F.5})$$

Then $\mathcal{E}(q_{R(B)}) \leq B$ and, with an absolute implicit constant,

$$R(B) = O\left(\log(d+2) + J(B)^2\right). \quad (\text{F.6})$$

Proof. Initially $\mathcal{E}(q_0) = \log(n!) \leq ndL$. While entropy exceeds n , Theorem F.2 contracts it by a factor at most $1 - 1/100 \leq e^{-1/100}$ per sweep. The first term in (F.5) therefore brings it to at most n . While it is between $n2^{-j}$ and $n2^{-(j-1)}$, the denominator in (F.3) is at most $100 + 12jL$. The j th summand suffices to halve the previous threshold, unless that threshold has already been crossed. The last dyadic threshold is at most B . Summing the affine-in- j ceilings proves (F.6). $\square$

Remark F.4 (What changed in the entropy proof). The exponent $3/2$ is subcritical for the endpoint-weight recursion and therefore gives a summed moment of order n . The scalar estimate also avoids a dimension-dependent split near zero square-root overlap. These features are what allow the entropy denominator used in the main argument to be logarithmic with constants independent of d .

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